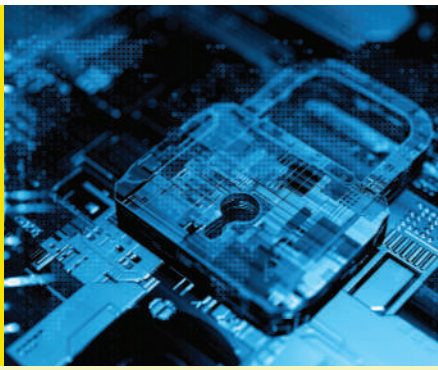
13

Security, Privacy and Ethical Issues in Information Systems



Principles

Policies and procedures must be established to avoid computer waste and mistakes.

Computer crime is a serious and rapidly growing area of concern requiring management attention.

Jobs, equipment and working conditions must be designed to avoid negative health effects.

Learning Objectives

- Describe some examples of waste and mistakes in an IS environment, their causes and possible solutions.
- Identify policies and procedures useful in eliminating waste and mistakes.
- Discuss the principles and limits of an individual's right to privacy.
- Explain the types and effects of computer crime.
- Identify specific measures to prevent computer crime.
- List the important effects of computers on the work environment.
- Identify specific actions that must be taken to ensure the health and safety of employees.
- Outline criteria for the ethical use of information systems.

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Why Learn About Security, Privacy and Ethical Issues in Information Systems?

Our last chapter will look at security, privacy and ethical issues, something that has been in the background throughout this book. A wide range of non-technical issues associated with the use of information systems provide both opportunities and threats to modern organizations. The issues span the full spectrum – from preventing computer waste and mistakes, to avoiding violations of privacy, to complying with laws on collecting data about customers, to monitoring employees. If you become a member of a human resources, information systems or legal department within an organization, you will likely be charged with leading the rest of the organization in dealing with these and other issues covered in this chapter. As a user of information systems, especially the Internet, it is in your own self-interest to become well versed on these issues. You need to know about the topics in this chapter to help avoid or recover from crime, fraud, privacy invasion or other potential problems.

13.1 Computer Waste and Mistakes

Computer-related waste and mistakes are major causes of computer problems, contributing as they do to unnecessarily high costs and lost profits. Computer waste involves the inappropriate use of computer technology and resources. It includes employees wasting computer resources and time by playing games and surfing the web, sending unnecessary email, printing documents and other material that is then not read, developing systems that are not used to their full extent, and discarding old hardware when it could be recycled or given to charity. UK-based Computers for Charities, for instance, will collect old technology, wipe clean any data stored on them and deliver them to charities where they are still useful. Junk email, also called spam, and junk faxes also cause waste. People receive hundreds of email messages and faxes advertising products and services not wanted or requested. Not only does this waste time, but it also wastes paper and computer resources. Worse still, spam messages often carry attached files with embedded viruses that can cause networks and computers to crash or allow hackers to gain unauthorized access to systems and data. Image-based spam is a new tactic spammers use to circumvent spam-filtering software that rejects email based on the content of messages and the use of keywords. The message is presented in a graphic form that can be read by people but not computers. This form of spam can be quite offensive and may contain pornographic photos and extremely graphic language.[1, 2] When waste is identified, it typically points to one common cause: the improper management of information systems and resources.

Computer-related mistakes refer to errors, failures and other computer problems that make computer output incorrect or not useful, caused mostly by human error. Despite many people's distrust, computers themselves rarely make mistakes. Even the most sophisticated hardware cannot produce meaningful output if users do not follow proper procedures. Mistakes can be caused by unclear expectations and a lack of feedback. Or a programmer might develop a program that contains errors. In other cases, a data-entry administrator might enter the wrong data. Unless errors are caught early and prevented, the speed of computers can intensify mistakes. As information technology becomes faster, more complex and more powerful, organizations and computer users face increased risks of experiencing the results of computer-related mistakes.

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Preventing Computer-Related Waste and Mistakes

To remain profitable in a competitive environment, organizations must use all resources wisely. Preventing computer-related waste and mistakes like those just described should therefore be a goal. To achieve it involves (1) establishing, (2) implementing, (3) monitoring and (4) reviewing effective policies and procedures.

Establishing Policies and Procedures

The first step to prevent computer-related waste is to establish policies and procedures regarding efficient acquisition, use and disposal of systems and devices. Most companies have implemented stringent policies on the acquisition of computer systems and equipment, including requiring a formal justification statement before computer equipment is purchased, the definition of standard computing platforms (operating system, type of computer chip, minimum amount of RAM, etc.) and the use of preferred vendors for all acquisitions.

Prevention of computer-related mistakes begins by identifying the most common types of errors, of which there are surprisingly few. Types of computer-related mistakes include the following:

- Data-entry or data-capture errors.
- Errors in computer programs.
- Errors in handling files, including formatting a disc by mistake, copying an old file over a newer one and deleting a file by mistake.
- Mishandling of computer output.
- Inadequate planning for and control of equipment malfunctions.
- Inadequate planning for and control of environmental difficulties (electrical problems, humidity problems, etc.).
- Installing computing capacity inadequate for the level of activity on corporate websites.
- Failure to provide access to the most current information by not adding new and not deleting old URL links.

Training programmes for individuals, and workgroups, and manuals and documents on how computer systems are to be maintained and used, can help prevent problems. Other preventative measures include needing approval for certain systems and applications before they are implemented and used to ensure compatibility and cost-effectiveness, and a requirement that documentation and descriptions of certain applications be submitted to a central office. This should include all cell formulas for spreadsheets and a description of all data elements and relationships in a database system (which, as we saw in Chapter 5, is already recorded in the data dictionary). After companies have planned and developed policies and procedures, they must consider how best to implement them.

Sometimes, computer error combines with human procedural errors to lead to the loss of human life. In March 2003, a Patriot missile battery on the Kuwait border accidentally shot down a British Royal Air Force Tornado GR-4 aircraft that was returning from a mission over Iraq. Two British pilots were killed in the incident. Many defence industry experts think the accident was caused by problems with the Patriot's radar combined with human error.

Implementing Policies and Procedures

Implementing policies and procedures to minimize waste and mistakes varies according to the type of business. Most companies develop such policies and procedures with advice from the firm's internal auditing group or its external auditing firm. The policies often focus

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on the implementation of source data automation, the use of data editing to ensure data accuracy and completeness, and the assignment of responsibility for data accuracy within each information system. Some useful policies to minimize waste and mistakes include the following:

- Changes to critical tables, HTML and URLs should be tightly controlled, with all changes authorized by responsible owners, and documented.
- A user manual should be available that covers operating procedures and documents the management and control of the application.
- Each system report should indicate its general content in its title and specify the time period it covers.
- The system should have controls to prevent invalid and unreasonable data entry.
- Controls should exist to ensure that data input, HTML and URLs are valid, applicable and posted in the right time frame.
- Users should implement proper procedures to ensure correct input data.

Training is another key aspect of implementation. Many users are not properly trained in using applications, and their mistakes can be very costly. One home in the small town of Valparaiso, in the USA, fairly valued at $88,550 was incorrectly recorded in the county's computer system as being worth over $290 million. The erroneous figure was used to forecast future income from property taxes. When the error was uncovered, the local school district and government agencies were forced to slash their budgets by €1.5 million when they found they wouldn't be getting the tax money after all.[3]

Monitoring Policies and Procedures

To ensure that users throughout an organization are following established procedures, the next step is to monitor routine practices and take corrective action if necessary. By understanding what is happening in day-to-day activities, organizations can make adjustments or develop new procedures. Many organizations implement internal audits to measure actual results against established goals, such as percentage of end-user reports produced on time, percentage of data-input errors detected, number of input transactions entered per eight-hour shift and so on.

Reviewing Policies and Procedures

The final step is to review existing policies and procedures and determine whether they are adequate. During review, people should ask the following questions:

- Do current policies cover existing practices adequately? Were any problems or opportunities uncovered during monitoring?
- Is the organization planning any new activities in the future? If so, does it need new policies or procedures on who will handle them and what must be done?
- Are contingencies and disasters covered?

This review and planning allows companies to take a proactive approach to problem solving, which can enhance a company's performance, such as by increasing productivity and improving customer service. Information systems professionals and users still need to be aware of the misuse of resources throughout an organization. Preventing errors and mistakes is one way to do so. Another is implementing in-house security measures and legal protections to detect and prevent a dangerous type of misuse: computer crime.

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Information Systems @ Work



Admiral to Price Car Insurance Based on Facebook Posts

To many, it's a terrifying idea. An insurance company peaks into private aspects of your life in order to help it set your premiums. In the past, a controversial example of this has been genetics. What if an insurance company had access to your genetic information, decided it was likely that you were going to get sick in the future, and refuse health cover? Maybe you might think this would be OK, but what if they predicted you would get sick in five years, covered you for four years and then refused to renew your policy? Would that be fair? Many people might see this as an 'insider-information' scenario and deem it unacceptable. Alternatively, people might agree to sharing personal data with an insurance company in the hope of getting lower premiums.

In November 2016, one of the biggest insurance companies in the UK, Admiral, announced that it was going to use data from social media to analyze the personalities of car owners and set the price of their insurance. Targeting first-time car owners, the scheme - called Firstcarquote – would have examined the Facebook posts of customers, with their permission, looking for personality traits that are correlated with safe driving. For example, conscientious and well-organized individuals might write in full sentences rather than in notes and might make arrangements with definite times and places rather than using vague notions. On the other hand, overconfident individuals might make use of exclamation marks and words like 'always' and 'never' rather than 'maybe'.

'It is incredibly transparent. If you don't want to use it in a quote then you don't have to', said Dan Mines, leader of the scheme. 'We are doing our best to build a product that allows young people to identify themselves as safe drivers. This is very much a test product for us. This is innovative, it is the first time anyone has done this. It is a test, this is early days. The data will only ever provide a discount. We will work through that and learn more.' Admiral estimated that first-time car owners, who pay on average around €2,000 for car insurance, could save up to 15 per cent via Firstcarquote.

Yossi Borenstein, the principal data scientist on Firstcarquote, told the *Guardian* newspaper that the scheme is based on algorithms developed by Admiral that will learn over time about the relationship between social media content and claims. 'Our analysis is not based on any one specific model, but rather on thousands of different combinations of likes, words and phrases and is constantly changing with new evidence that we obtain from the data', he said. 'As such our calculations reflect how drivers generally behave on social media, and how predictive that is, as opposed to fixed assumptions about what a safe driver may look like.' He also stressed that Admiral would not have access to information about what its customers look at on Facebook or what their friends do. The company would only have access to the information gathered during the quote process and would have no ongoing access.

However, hours before the scheme was set to launch, Facebook stepped on the brakes. They said that protecting the privacy of its users was of the utmost importance to it and, as a result, did not allow Firstcarquote to proceed. Facebook's terms and conditions state that its data should not be used to 'make decisions about eligibility, including whether to approve or reject an application or how much interest to charge on a loan'. Mines said, 'People share their social data with third parties every day, without any financial benefit to themselves. Firstcarquote asks users to share their data only once and offers them a clear, tangible financial benefit in return.'

Questions

1. In what ways does your personality 'shine through' in your social media posts?
2. Who should 'own' the content that you publish on social media?
3. Should insurance companies be able to use social media posts to help set premiums?
4. What protections for users should be in place from companies using their social media data?

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13.2 Computer Crime

According to Financial Fraud Action UK, in 2012 credit card fraudsters stole nearly €400 million in the UK alone, with €163 million of this happening through e-commerce transactions.[4] In 2013, Internet security experts Kaspersky Lab detected almost 3 billion malware attacks on user computers and 104,427 new malicious programs aimed at mobile devices.[5] The term computer crime covers a wide variety of activities, including these. Some more examples are listed next and then some types of computer crime are discussed.

- The largest consumer fraud in the USA was committed by the Gambino crime family involving two different computer-related ploys and resulted in a loss to the public of over €200 million. One of the schemes offered 'free' tours of adult Internet sites but required the victim to provide a credit card supposedly for age-verification purposes. Victims took the free tours and then their credit cards were hit for charges over and over again. The second prong to this scheme involved the use of a third-party billing provider to add charges on people's telephone bills for services not provided.[6]
- A 20-year-old man was sentenced to 57 months in prison for hijacking more than 400,000 PCs over the Internet and turning them into a 'botnet' or 'zombie network', a network of personal computers used to perform a task without the owner's knowledge. He would then rent the zombie network out to spyware distributors, hackers and spammers to use in performing their work.[7]
- Russian organized crime extorted untold thousands of dollars from firms doing business on the Internet by demanding €7,000 or more for protection from being hit by a denial-of-service attack on their website. Some firms bought the 'protection'; some of those that did not were attacked.[8]
- A British information systems expert accessed a series of computer networks used by the US Army, Navy, Air Force and Department of Defense, searching for what he called 'suppressed technology'. US authorities claimed he caused more than $700,000 of damage.[9]
- The UK government's tax credit website, which allowed qualifying citizens to claim tax benefits, was shut down in 2005 because it was being targeted by organized gangs claiming many millions of pounds from the government.[10]

Online Fraud
There is no ultimate solution to prevent computer fraud. Fraudsters and security experts are locked in an endless game of cat and mouse. Battling them involves a constantly evolving mix of technical and social measures.



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Identity Theft

Identity theft is one of the fastest growing crimes. It is a crime where an imposter obtains key pieces of personal identification information, such as date of birth, address, national insurance number and mother's maiden name, and uses them to open bank accounts, get credit cards, loans, benefits and documents such as passports and driving licences in the victim's name. In other cases, the identity thief uses personal information to gain access to the person's existing accounts. Typically, the thief changes the mailing address on an account and runs up a huge bill before the person whose identity has been stolen realizes there is a problem. The Internet has made it easier for an identity thief to use the stolen information because transactions can be made without any personal interaction. The UK government has a website, nidirect.gov.uk/articles/protect-your-identity, to advise its citizens and help victims. A wide range of methods are used by the perpetrators of these crimes, which makes investigating them difficult. Frequently, a critical computer password has been talked out of a person, or guessed based on a knowledge of the person, a practice called **social engineering**. For example, many people use the name of their pet as their password. Many teenagers use the name of their favourite pop artist. Alternatively, the attackers might simply go through the person's rubbish, looking for a discarded utility bill or bank statement. In addition, over 2,000 websites offer the digital tools – for free – that will let people snoop, crash computers, hijack control of a machine or retrieve a copy of every keystroke.

social engineering Using one's social skills to get computer users to provide you with information to access an information system or its data.

Another popular method to get information is 'shoulder surfing' – the identity thief simply stands next to someone at a public office, such as the passport office or even when filling in a form to join a customer loyalty programme, and watches as the person fills out personal information on a form. The same thing can happen at a bank ATM where the attacker simply watches the person enter their PIN, or at a shop when the victim is using their credit card to make a purchase (see Figure 13.1).



Figure 13.1 Shoulder Surfing *Always take care when using an ATM that no one can see you enter your PIN.*

Consumers can help protect themselves by regularly checking their credit reports, following up with creditors if their bills do not arrive on time, not revealing any personal information in response to unsolicited email or phone calls, and shredding bills and other documents that contain sensitive information.[11]

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Cyberterrorism

Government officials and IS security specialists have documented a significant increase in Internet probes and server scans since early 2001. A growing concern among authorities is that such intrusions are part of an organized effort by cyberterrorists to map potential security holes in critical systems. A **cyberterrorist** is someone who intimidates or coerces a government or organization to advance their political or social objectives by launching computer-based attacks against computers, networks and the information stored on them. Attacks would likely be aimed at critical infrastructure, which includes telecommunications, energy, banking and finance, water systems, government operations and emergency services. Successful cyberattacks against the facilities that provide these services could cause widespread and massive disruptions to the normal functioning of a society.

cyberterrorist Someone who intimidates or coerces a government or organization to advance his or her political or social objectives by launching computer-based attacks against computers, networks and the information stored on them.

A similar term, 'cyberwar', is arguably not a crime but involves a country or state attacking another, using the same techniques as a cyberterrorist.

Illegal Access and Use

Crimes involving illegal system access and use of computer services are a concern to both government and business. Since the outset of information technology, computers have been plagued by criminal crackers. A **cracker**, often called a **hacker**, although this term has a range of meanings, is a computer-savvy person who attempts to gain unauthorized or illegal access to computer systems. Often they are 'just looking' but could also be trying to corrupt files, steal data or even transfer money. In many cases, crackers are people who are looking for fun and excitement – the challenge of beating the system. **Script kiddies** admire crackers, but have little technical savvy. They are crackers who download programs called 'scripts' that automate the job of breaking into computers. **Insiders** are employees, disgruntled or otherwise, working solo or in concert with outsiders to compromise corporate systems.

cracker or **hacker** A computer-savvy person who attempts to gain unauthorized or illegal access to computer systems.

script kiddies A cracker with little technical savvy who downloads programs called scripts, which automate the job of breaking into computers.

insiders An employee, disgruntled or otherwise, working solo or in concert with outsiders to compromise corporate systems.

Catching and convicting criminal hackers remains a difficult task. The method behind these crimes is often hard to determine. Even if the method behind the crime is known, tracking down the criminals can take a lot of time.

Data and information are valuable corporate assets. The intentional use of illegal and destructive programs to alter or destroy data is as much a crime as destroying tangible goods. The most common of these programs are viruses and worms, which are software programs that when loaded into a computer system, will destroy, interrupt or cause errors in processing. Such programs are also called 'malware'.

A **virus** is a computer program file capable of attaching to discs or other files and replicating itself repeatedly, typically without the user's knowledge or permission. Some viruses attach to files, so when the infected file executes, the virus also executes. Other viruses sit in a computer's memory and infect files as the computer opens, modifies or creates the files. They are often disguised as games or images with clever or attention-grabbing titles such as 'Boss, naked'. Some viruses display symptoms, and some viruses damage files and computer systems. The m00p virus gang, for example, conspired to infect computers with a virus that would turn each infected machine into a zombie machine under their control. The zombie network could then be used to spread viruses and other malware across the Internet, without the owners of the compromised computers even being aware.[12] Hoax viruses can also be a problem. A hoax virus is a message, usually distributed by email, warning recipients to carry out a procedure on their computer to protect themselves from a 'virus threat', when the procedure itself is actually doing the damage. Typically a hoax virus encourages people to delete an important systems file. The message will encourage people to forward it on to all their contacts.

virus A computer program file capable of attaching to discs or other files and replicating itself repeatedly, typically without the user's knowledge or permission.

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Worms are computer programs that replicate but, unlike viruses, do not infect other computer program files. Worms can create copies on the same computer or can send the copies to other computers via a network. Worms often spread via Internet Relay Chat (IRC). For example, the MyDoom worm, also known as Shimgapi and Novarg, started spreading in January 2004 and quickly became the most virulent email worm ever. The worm arrived as an email with an attachment with various names and extensions, including .exe, .scr, .zip and .pif. When the attachment executed, the worm sent copies of itself to other email addresses stored in the infected computer. The first version of the virus, MyDoom.A, was designed to attack The SCO Group Inc.'s website. A later variant, dubbed MyDoom.B, was designed to enable similar denial-of-service attacks against the Microsoft website. The B variant also included a particularly nasty feature in that it blocked infected computers from accessing sites belonging to vendors of antivirus products. Infected email messages carrying the MyDoom worm have been intercepted from over 142 countries and at one time accounted for 1 in every 12 email messages.

worm A parasitic computer program that can create copies of itself on the infected computer or send copies to other computers via a network.

A **Trojan horse** program is a malicious program that disguises itself as a useful application and purposefully does something the user does not expect. Trojans are not viruses because they do not replicate, but they can be just as destructive. Many people use the term to refer only to non-replicating malicious programs, thus making a distinction between Trojans and viruses. A German language email, for example, was used to spread a Trojan horse that steals passwords and logon details of customers' online bank accounts and then relays them back to a remote server. The malware tried to get users to install the Trojan horse by disguising itself as a software patch for a new flaw in Microsoft software.[13] Spyware is often spread using the Trojan horse method. Spyware is software which records all manner of personal information about users and forwards it to the spyware's owner, all without the user's consent. Name, address, credit card numbers and passwords can all be collected by spyware, as can information on web browsing behaviour, which would be valuable for marketing.

Trojan horse A malicious program that disguises itself as a useful application and purposefully does something the user does not expect.

A logic bomb is a type of Trojan horse that executes when specific conditions occur. Triggers for logic bombs can include a change in a file by a particular series of keystrokes or at a specific time or date.

A variant is a modified version of a virus that is produced by the virus's author or another person who amends the original virus code. If changes are small, most antivirus products will also detect variants. However, if the changes are significant, the variant might go undetected by antivirus software.

In some cases, a virus or a worm can completely halt the operation of a computer system or network for days or longer until the problem is found and repaired. In other cases, a virus or a worm can destroy important data and programs. If back-ups are inadequate, the data and programs might never be fully functional again. The costs include the effort required to identify and neutralize the virus or worm and to restore computer files and data, as well as the value of business lost because of unscheduled computer downtime.

As a result of the increasing threat of viruses and worms, most computer users and organizations have installed **antivirus programs** on their computers. Such software runs in the background to protect your computer from dangers lurking on the Internet and other possible sources of infected files. Some antivirus software is even capable of repairing common virus infections automatically, without interrupting your work. The latest virus definitions are downloaded automatically when you connect to the Internet, ensuring that your PC's protection is current. To safeguard your PC and prevent it from spreading viruses to your friends and coworkers, some antivirus software scans and cleans both incoming and outgoing email messages. Table 13.1 lists some of the most popular antivirus software.

antivirus program Software that runs in the background to protect your computer from dangers lurking on the Internet and other possible sources of infected files.

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Table 13.1 Antivirus Software

Antivirus Software	Software Manufacturer	Website
Symantec's Norton AntiVirus	Symantec	www.broadcom.com
McAfee Virus Scan	McAfee	www.mcafee.com
Panda Antivirus Platinum	Panda Software	www.pandasoftware.com
Sophos Antivirus	Sophos	www.sophos.com
PC-cillin	Trend Micro	www.trendmicro.com

Proper use of antivirus software requires the following steps:

1. Install antivirus software. These programs should automatically check for viruses each time you boot up your computer or insert a flash drive or CD, and some even monitor all email and file transmissions and copying operations.
2. Ensure the antivirus software updates often. New viruses are created all the time, and antivirus software suppliers are constantly updating their software to detect and take action against these new viruses. The software should itself check for updates regularly, without the need for an instruction from the user.
3. Scan all removable media, including CDs, before copying or running programs from them. Hiding on flash drives or CDs, viruses often move between systems. If you carry document or program files on removable media between computers at school or work and your home system, always scan them.
4. Install software only from a sealed package or secure website of a known software company. Even software publishers can unknowingly distribute viruses on their program discs or software downloads. Most scan their own systems, but viruses might still remain.
5. Follow careful downloading practices. If you download software from the Internet or a bulletin board, check your computer for viruses immediately after completing the transmission.
6. If you detect a virus, take immediate action. Early detection often allows you to remove a virus before it does any serious damage.

Despite careful precautions, viruses can still cause problems. They can elude virus-scanning software by lurking almost anywhere in a system. Future antivirus programs might incorporate 'nature-based models' that check for unusual or unfamiliar computer code. The advantage of this type of virus program is the ability to detect new viruses that are not part of an antivirus database.

Hoax, or false, viruses are another problem. Crackers sometimes warn the public of a new and devastating virus that doesn't actually exist just to create fear. Companies sometimes spend hundreds of hours warning employees and taking preventative action against a non-existent virus. Security specialists recommend that IS personnel establish a formal paranoia policy to thwart virus panic among gullible end-users. Such policies should stress that before users forward an email alert to colleagues, they should send it to the help desk or the security team. The corporate intranet can be used to explain the difference between real viruses and fakes, and it can provide links to websites to set the record straight.

Be aware that virus writers also use known hoaxes to their advantage. For example, AOL4FREE began as a hoax virus warning. Then, a hacker distributed a destructive Trojan attached to the original hoax virus warning. Always remain vigilant and never open a suspicious attachment.

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Equipment Theft

During illegal access to computer systems, data can be stolen. In addition to theft of data and software, all types of computer systems and equipment have been stolen from offices. Mobile computers such as laptops and smartphones are especially easy for thieves to take. Very often the data stored on these devices is more valuable than the device itself. An MI5 agent's laptop containing sensitive government information was stolen at Paddington train station in London, and a senior British Army official's laptop was taken at Heathrow Airport.[14] To fight computer crime, many companies use devices that disable the disc drive and/or lock the computer to the desk.

Software and Internet Software Piracy

Like books and movies – and other intellectual properties – software is protected by copyright laws. Often, people who would never think of plagiarizing another author's written work have no qualms about using and copying software programs they have not paid for. Such illegal duplicators are called 'pirates'; the act of illegally duplicating software is called **software piracy**.

software piracy The act of illegally duplicating software.

Technically, software purchasers are granted the right only to use the software under certain conditions; they don't really own the software. Licences vary from program to program and can authorize as few as one computer or one person to use the software or as many as several hundred network users to share the application across the system. Making additional copies or loading the software onto more than one machine might violate copyright law and be considered piracy.

The Business Software Alliance estimates that the software industry loses over €8 billion per year in revenue to software piracy annually. Half the loss comes from Asia, where China and Indonesia are the biggest offenders. In Western Europe, annual piracy losses range between €1.5 and €2 billion. Although the rate of software piracy is quite high in Latin America and Central Europe, those software markets are so small that the monetary losses are considerably lower. Overall, it is estimated that 35 per cent of the world's software is pirated.[15]

Internet-based software piracy occurs when software is illegally downloaded from the Internet. It is the most rapidly expanding type of software piracy and the most difficult form to combat. The same purchasing rules apply to online software purchases as for traditional purchases. Internet piracy can take several forms, including the following:

- Pirate websites that make software available for free or in exchange for uploaded programs.
- Internet auction sites that offer counterfeit software, which infringes copyrights.
- Peer-to-peer networks, which enable unauthorized transfer of copyrighted programs.

Computer-Related Scams

People have lost hundreds of thousands of euros on property, travel, stock and other business scams. Now, many of these scams are being perpetrated with computers. Using the Internet, scam artists offer get-rich-quick schemes involving bogus property deals, tout 'free' holidays with huge hidden costs, commit bank fraud, offer fake telephone lotteries, sell worthless penny stocks and promote illegal tax-avoidance schemes.

Over the past few years, credit card customers of various banks have been targeted by scam artists trying to get personal information needed to use their credit cards. The scam typically works by sending an email to many thousands of people, asking them to click on a link that seems to direct users to a bank's website to fill in essential security information. Some of the recipients will probably be customers of the bank. At the site, they are asked for their full debit and credit card numbers and expiration dates, their name, address and other personal information. The problem is that the website customers are directed to is a fake site operated by someone

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trying to gain access to that information. This form of scam is called 'phishing'. The website used is often extremely similar to the bank's real website and may contain links to the real site.

In the weeks following Hurricane Katrina in the USA, the FBI warned that over half the Hurricane Katrina aid sites it checked were registered to people outside the USA and likely to be fraudulent. A 20-year-old man was charged with setting up websites designed to look like those of the American Red Cross and other organizations accepting donations to help the victims. He then sold these to 'would-be scammers' for about $140 each. For his trouble, this person faced 50 years in prison and a fine of $1 million.[16]

The following is a list of tips to help you avoid becoming a scam victim:

- Don't agree to anything in a high-pressure meeting or seminar. Insist on having time to think it over and to discuss things with your spouse, your partner or even your solicitor. If a company won't give you the time you need to check it out and think things over, you don't want to do business with them. A good deal now will be a good deal tomorrow; the only reason for rushing you is if the company has something to hide.
- Don't judge a company based on appearances. Professional-looking websites can be created and published in a matter of days. After a few weeks of taking money, a site can vanish without a trace in just a few minutes. You might find that the perfect money-making opportunity offered on a website was a money-maker for the crook and a money-loser for you.
- Avoid any plan that pays commissions simply for recruiting additional distributors. Your primary source of income should be your own product sales. If the earnings are not made primarily by sales of goods or services to consumers or sales by distributors under you, you might be dealing with an illegal pyramid.
- Beware of 'shills', people paid by a company to lie about how much they've earned and how easy the plan was to operate. Check with an independent source to make sure that you aren't having the wool pulled over your eyes.
- Beware of a company's claim that it can set you up in a profitable home-based business but that you must first pay up front to attend a seminar and buy expensive materials. Frequently, seminars are high-pressure sales pitches, and the material is so general that it is worthless.
- If you are interested in starting a home-based business, get a complete description of the work involved before you send any money. You might find that what you are asked to do after you pay is far different from what was stated in the ad. You should never have to pay for a job description or for needed materials.
- Get in writing the refund, buy-back and cancellation policies of any company you deal with. Do not depend on oral promises.
- If you need advice about an online solicitation or if you want to report a possible scam, contact your country's computer crime unit. In the UK, you can find more information at www.direct.gov.uk.

13.3 Preventing Computer-Related Crime

Because of increased computer use, greater emphasis is placed on the prevention and detection of computer crime. Many countries have passed data laws governing how data can be stored, processed and transferred, and laws on computer crime. Some believe that these laws are not effective because companies do not always actively detect and pursue computer crime, security is inadequate and convicted criminals are not severely punished. However, all

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over the world, private users, companies, employees and public officials are making individual and group efforts to curb computer crime, and recent efforts have met with some success.

Crime Prevention by the State

In the UK, the Computer Misuse Act of 1990, which criminalizes unauthorized access to computer systems, and the Data Protection Act of 2018, which governs when and how data about individuals can be stored and processed, have been passed. Many countries have passed similar laws.

In the UK, the Home Office is charged with tackling computer crime with some police forces having a 'cyber crime' unit. The Information Commissioner's Office is in charge of the UK's independent authority set up to protect personal information (and as we shall see later in this chapter, to promote access to official information). The UK also has an organization dedicated to fighting specific types of computer crime. The Child Exploitation and Online Protection Centre (CEOP) tackles child sex abuse, especially where it has been facilitated in some way by the Internet.

Crime Prevention by Organizations

Companies are also taking crime-fighting efforts seriously. Many businesses have designed procedures and specialized hardware and software to protect their corporate data and systems. Specialized hardware and software, such as encryption devices, can be used to encode data and information to help prevent unauthorized use. Encryption is the process of converting an original electronic message into a form that can be understood only by the intended recipients. A key is a variable value that is applied using an algorithm to a string or block of unencrypted text to produce encrypted text or to decrypt encrypted text. Encryption methods rely on the limitations of computing power for their effectiveness – if breaking a code requires too much computing power, even the most determined code crackers will not be successful. The length of the key used to encode and decode messages determines the strength of the encryption algorithm.

Public-key infrastructure (PKI) enables users of an unsecured public network such as the Internet to securely and privately exchange data through the use of a public and a private cryptographic key pair that is obtained and shared through a trusted authority. PKI is the most common method on the Internet for authenticating a message sender or encrypting a message. PKI uses two keys to encode and decode messages. One key of the pair, the message receiver's public key, is readily available to the public and is used by anyone to send that individual encrypted messages. The second key, the message receiver's private key, is kept secret and is known only by the message receiver. Its owner uses the private key to decrypt messages – convert encoded messages back into the original message. Knowing a person's public key does not enable you to decrypt an encoded message to that person.

public-key infrastructure (PKI) A means to enable users of an unsecured public network such as the Internet to securely and privately exchange data through the use of a public and a private cryptographic key pair that is obtained and shared through a trusted authority.

Using **biometrics** is another way to protect important data and information systems. Biometrics involves the measurement of one of a person's traits, whether physical or behavioural. Biometric techniques compare a person's unique characteristics against a stored set to detect differences between them. Biometric systems can scan fingerprints, faces, handprints, irises and retinal images to prevent unauthorized access to important data and computer resources. Most of the interest among corporate users is in fingerprint technology, followed by face recognition. Fingerprint scans hit the middle ground between price and effectiveness (see Figure 13.2). Iris and retina scans are more accurate, but they are more expensive and involve more equipment.

biometrics The measurement of one of a person's traits, whether physical or behavioural.

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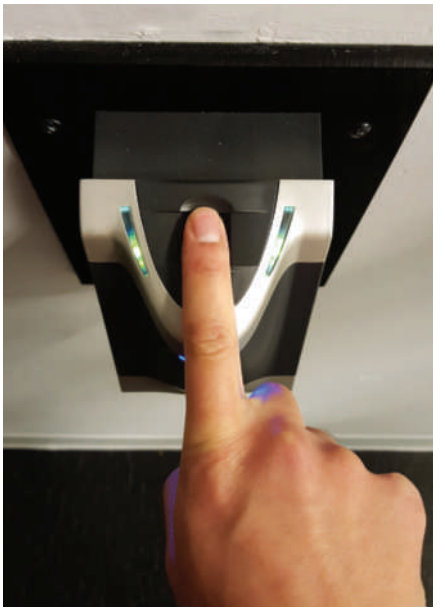


Figure 13.2 Fingerprint Authentication *Fingerprint authentication devices provide security in the PC environment by using fingerprint information instead of passwords.*

As employees move from one position to another at a company, they can build up access to multiple systems if inadequate security procedures fail to revoke access privileges. It is clearly not appropriate for people who have changed positions and responsibilities to still have access to systems they no longer use. To avoid this problem, many organizations create role-based system access lists so that only people filling a particular role (e.g. line manager) can access a specific system.

Crime-fighting procedures usually require additional controls on the information system. Before designing and implementing controls, organizations must consider the types of computer-related crime that might occur, the consequences of these crimes, and the cost and complexity of needed controls. In most cases, organizations conclude that the trade-off between crime and the additional cost and complexity weighs in favour of better system controls. Having knowledge of some of the methods used to commit crime is also helpful in preventing, detecting and developing systems resistant to computer crime (see Table 13.2). Some companies actually hire former criminals to thwart other criminals.

Although the number of potential computer crimes appears to be limitless, the actual methods used to commit crime are limited. The following list provides a set of useful guidelines to protect your computer from criminal hackers:

- Install strong user authentication and encryption capabilities on your firewall.
- Install the latest security patches, which are often available at the vendor's Internet site.
- Disable guest accounts and null user accounts that let intruders access the network without a password.
- Do not provide overfriendly logon procedures for remote users (e.g. an organization that used the word 'welcome' on their initial logon screen found they had difficulty prosecuting a criminal hacker).
- Restrict physical access to the server and configure it so that breaking into one server won't compromise the whole network.

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- Give each application (email, FTP and domain name server) its own dedicated server.
- Turn audit trails on.
- Consider using caller ID.
- Install a corporate firewall between your corporate network and the Internet.
- Install antivirus software on all computers and regularly download vendor updates.
- Conduct regular IS security audits.
- Verify and exercise frequent data back-ups for critical data.

Table 13.2 Common Methods Used to Commit Computer Crimes

Methods	Examples
Add, delete or change inputs to the computer system	Delete records of absences from class in a student's school records
Modify or develop computer programs that commit the crime	Change a bank's program for calculating interest to make it deposit rounded amounts in the criminal's account
Alter or modify the data files used by the computer system	Change a student's grade from C to A
Operate the computer system in such a way as to commit computer crime	Access a restricted government computer system
Divert or misuse valid output from the computer system	Steal discarded printouts of customer records from a company trash bin
Steal computer resources, including hardware, software and time on computer equipment	Make illegal copies of a software program without paying for its use
Offer worthless products for sale over the Internet	Send email requesting money for worthless hair growth product
Blackmail executives to prevent release of harmful information	Eavesdrop on organization's wireless network to capture competitive data or scandalous information
Blackmail company to prevent loss of computer-based information	Plant logic bomb and send letter threatening to set it off unless paid considerable sum

Companies are also joining together to fight crime. The Software and Information Industry Alliance (SIIA) was the original antipiracy organization, formed and financed by many of the large software publishers. Microsoft financed the formation of a second antipiracy organization, the Business Software Alliance (BSA). The BSA, through intense publicity, has become the more prominent organization. Other software companies, including Apple, Adobe, Hewlett-Packard and IBM, now contribute to the BSA.

Crime Prevention by Individuals

A number of individuals – victims, former criminals, concerned parents – have set up websites offering support for those worried about computer crime, and advice on how to fight it.

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Using Intrusion Detection Software

intrusion detection system (IDS) Software that monitors system and network resources and notifies network security personnel when it senses a possible intrusion.

An **intrusion detection system (IDS)** monitors system and network resources and notifies network security personnel when it senses a possible intrusion. Examples of suspicious activities include repeated failed logon attempts, attempts to download a program to a server and access to a system at unusual hours. Such activities generate alarms that are captured on log files. Intrusion detection systems send an alarm, often by email or pager, to network security personnel when they detect an apparent attack. Unfortunately, many IDSs frequently provide false alarms that result in wasted effort. If the attack is real, network security personnel must make a decision about what to do to resist the attack. Any delay in response increases the probability of damage from a criminal hacker attack. Use of an IDS provides another layer of protection in the event that an intruder gets past the outer security layers – passwords, security procedures and corporate firewall.

The following story is true, but the company's name has been changed to protect its identity. The ABC company employs more than 25 IDS sensors across its worldwide network, enabling it to monitor 90 per cent of the company's internal network traffic. The remaining 10 per cent comes from its engineering labs and remote sales offices, which are not monitored because of a lack of resources. The company's IDS worked very well in providing an early warning of an impending SQL Slammer attack. The Slammer worm had entered the network via a server in one of the engineering labs. The person monitoring the IDS noticed outbound traffic consistent with SQL Slammer at about 7.30 am. He contacted the network operations group by email and followed up with a phone call and a voice mail message. Unfortunately, the operations group gets so many emails that if a message is not highlighted as URGENT, the message might be missed. That is exactly what happened – the email alert wasn't read and the voice message wasn't retrieved in time to block the attack. A few hours later, the ABC company found itself dealing with a massive number of reports of network and server problems.

Using Managed Security Service Providers (MSSPs)

Keeping up with computer criminals – and with new regulations – can be daunting for organizations. Criminal hackers are constantly poking and prodding, trying to breach the security defences of companies. For most small and mid-sized organizations, the level of in-house network security expertise needed to protect their business operations can be quite costly to acquire and maintain. As a result, many are outsourcing their network security operations to managed security service providers (MSSPs) such as Counterpane, Guardent, Internet Security Services, Riptech and Symantec. MSSPs monitor, manage and maintain network security for both hardware and software. These companies provide a valuable service for IS departments drowning in reams of alerts and false alarms coming from virtual private networks (VPNs); antivirus, firewall and intrusion detection systems; and other security monitoring systems. In addition, some provide vulnerability scanning and web blocking/filtering capabilities.

Preventing Crime on the Internet

As mentioned in Chapter 6, Internet security can include firewalls and many methods to secure financial transactions. A firewall can include both hardware and software that act as a barrier between an organization's information system and the outside world. Some systems have been developed to safeguard financial transactions on the Internet.

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To help prevent crime on the Internet, the following steps can be taken:

1. Develop effective Internet usage and security policies for all employees.
2. Use a stand-alone firewall (hardware and software) with network monitoring capabilities.
3. Deploy intrusion detection systems, monitor them and follow up on their alarms.
4. Monitor managers and employees to make sure that they are using the Internet for business purposes.
5. Use Internet security specialists to perform audits of all Internet and network activities.

Even with these precautions, computers and networks can never be completely protected against crime. One of the biggest threats is from employees. Although firewalls provide good perimeter control to prevent crime from the outside, procedures and protection measures are needed to protect against computer crime by employees. Passwords, identification numbers and tighter control of employees and managers also help prevent Internet-related crime.

13.4 Privacy

Privacy is a big issue for many people. When information is computerized and can be processed and transferred easily, augmented and collated, summarized and reported, privacy concerns grow. The European Union has a data-protection directive that requires firms transporting data across national boundaries to have certain privacy procedures in place. This directive affects virtually any company doing business in Europe, and it is driving much of the attention being given to privacy in the USA.

Privacy and the Government

Many people are suspicious of the government when it comes to information that is stored about them. In the UK a few years ago, the government wanted an identity card scheme which, it was claimed, would help fight international terrorism and identity theft and other fraud. The card would be linked to a database, which would hold names, addresses and biometric information on all citzens. Expected to cost many billions of euros, some people pledged never to carry them, claiming that the scheme would create a 'big brother' society. Many of these fears were unfounded, although the debate did highlight a lack of trust in the state and the scheme was not implemented.

Many governments are in fact quite open about the information that they store. Numerous countries have implemented some sort of freedom of information legislation. In South Africa, it is the Promotion of Access to Information Act. In the UK it is the Freedom of Information Act. Similar laws have been passed throughout Europe.

The UK Freedom of Information Act governs all data that is not about an individual, in any public organization including government, local councils, schools, universities and hospitals. The Act basically states that all such organizations must give out whatever information is requested of them, as long as it is not about an individual (which is protected under the Data Protection Act) or some other sensitive information. So, for example, you would be able to ask your university how many people achieved A grades in one of your modules last year (this information is probably published on the students' portal anyway). However, you couldn't request information about a professor's salary. You could, though, ask for information about lecturers' pay scales (which again is already freely available from the relevant union's website).

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Privacy at Work

The right to privacy at work is an important issue. Currently, the rights of workers who want their privacy and the interests of companies that demand to know more about their employees are in conflict. Recently, companies that have been monitoring their workers have raised concerns. For example, workers might find that they are being closely monitored via computer technology. These computer-monitoring systems tie directly into workstations; specialized computer programs can track every keystroke made by a user. This type of system can determine what workers are doing while at the keyboard. The system also knows when the worker is not using the keyboard or computer system. These systems can estimate what people are doing and how many breaks they are taking. Needless to say, many workers consider this close supervision very dehumanizing.

Email Privacy

Email also raises some interesting issues about work privacy. A company has the right to look at any data stored on its servers, which includes its email servers and therefore all messages sent by or to its employees. Many companies routinely store all emails sent or received for several years and many employees have lost their jobs for forwarding inappropriate messages. Others have sent embarassing messages that have been forwarded exponentially by recipients who pass the 'joke' on to their friends. Accidentally hitting 'reply all' to an email instead of 'reply' is such a common problem it's got its own thread on Reddit.[17]

Privacy and the Internet

Some people assume that there is no privacy on the Internet and that you use it at your own risk. Others believe that companies with websites should have strict privacy procedures and be accountable for privacy invasion. Regardless of your view, the potential for privacy invasion on the Internet is huge. People wanting to invade your privacy could be anyone from criminal hackers to marketing companies to corporate bosses. Email is a prime target, as discussed previously. When you visit a website, information about you and your computer can be captured. When this information is combined with other information, companies can know what you read, what products you buy and what your interests are. According to an executive of an Internet software monitoring company, 'It's a marketing person's dream'.

Most people who buy products on the web say it's very important for a site to have a policy explaining how personal information is used, and the policy statement must make people feel comfortable and be extremely clear about what information is collected and what will and will not be done with it. However, many websites still do not prominently display their privacy policy or implement practices completely consistent with that policy. The real issue that Internet users need to be concerned with is 'What do content providers want with our personal information?' If a site requests that you provide your name and address, you have every right to know why and what will be done with it. If you buy something and provide a shipping address, will it be sold to other retailers? Will your email address be sold on a list of active Internet shoppers? And, if so, you should realize that it's no different from the lists compiled from the orders you place with catalogue retailers – you have the right to be taken off any mailing list.

These same questions can be asked of Internet chat rooms that require you to register before you can post messages. It is important for the forum moderators to know who is posting, but users should also have confidence that their information will not be misused.

Platform for Privacy Preferences (P3P) A screening technology that shields users from websites that don't provide the level of privacy protection they desire.

A potential solution to some consumer privacy concerns is the screening technology called the **Platform for Privacy Preferences (P3P)** being proposed to shield users from sites that don't provide the level of privacy protection they desire. Instead of forcing users to find and read

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through the privacy policy for each site they visit, P3P software in a computer's browser will download the privacy policy from each site, scan it and notify the user if the policy does not match his or her preferences. (Of course, unethical marketers can post a privacy policy that does not accurately reflect the manner in which the data is treated.) The World Wide Web Consortium (W3C), an international industry group whose members include Apple, Commerce One, Ericsson and Microsoft, is supporting the development of P3P.

A social network service employs the web and software to connect people for whatever purpose. There are thousands of such networks, which have become popular among teenagers. Some of the more popular social networking websites include Tumblr, QZone, QQ, Viber, LinkedIn and Snapchat. Most of these allow one to easily create a user profile that provides personal details, photos, even videos that can be viewed by other visitors to the website. Some of the websites have age restrictions or require that a parent register their pre-teen by providing a credit card to validate the parent's identity. Teens can provide information about where they live, go to school, their favourite music and interests in the hopes of meeting new friends. Unfortunately, they can also meet ill-intentioned strangers at these sites. Many documented encounters involve adults masquerading as teens attempting to meet young people for illicit purposes. Parents are advised to discuss potential dangers, check their children's profiles and monitor their activities at such websites.

Whenever someone registers a domain name such as www.mydomain.co.uk, the name and address given during registration become public information and can be seen by simply running a 'whois' query, which can be easily done on many websites. Parents should be aware of this before they let their children have their own web page.

Fairness in Information Use

Selling information to other companies can be so lucrative that many companies will continue to store and sell the data they collect on customers, employees and others. When is this information storage and use fair and reasonable to the people whose data is stored and sold? Do people have a right to know about data stored about them and to decide what data is stored and used? As shown in Table 13.3, these questions can be broken down into four issues that should be addressed: knowledge, control, notice and consent.

Table 13.3 The Right to Know and the Ability to Decide

Fairness Issues	Database Storage	Database Usage
The right to know	Knowledge	Notice
The ability to decide	Control	Consent
Knowledge: Should people know what data is stored about them? In some cases, people are informed that information about them is stored in a corporate database. In others, they do not know that their personal information is stored in corporate databases		
Control: Should people be able to correct errors in corporate database systems? This is possible with most organizations, although it can be difficult in some cases		
Notice: Should an organization that uses personal data for a purpose other than the original purpose notify individuals in advance? Many companies don't do this even though they should.		
Consent: If information on people is to be used for other purposes, should these people be asked to give their consent before data on them is used? Many companies do not give people the ability to decide if information on them will be sold or used for other purposes		

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In Europe, the General Data Protection Regulation, discussed in the next section, governs the answers to these questions. The Regulation relates to data about individuals and states that:

1. Personal data shall be processed fairly and lawfully.
2. Companies must have a reason for collecting and storing the data – they cannot arbitrarily start hoarding it, and they cannot process it in any manner incompatible with that reason.
3. The data collected shall be adequate, relevant and not excessive in relation to the reason for collecting it.
4. Companies must make an effort to ensure the data is accurate and, where necessary, up to date.
5. The data will not be stored for longer than is necessary.
6. All of the above applies to processing the data, not just collecting and storing it.
7. Companies must take steps to ensure that the data is secure.
8. The data must not be transferred to somewhere that does not have a similar law on processing it.

The Regulation allows individuals to access information stored about them and, if necessary, have the data updated or deleted. Similar laws have been implemented throughout Europe.

Even though privacy laws for private organizations are not very restrictive, most organizations are very sensitive to privacy issues and fairness. They realize that invasions of privacy can hurt their business, turn away customers and dramatically reduce revenues and profits. Consider a major international credit card company. If the company sold confidential financial information on millions of customers to other companies, the results could be disastrous. In a matter of days, the firm's business and revenues could be reduced dramatically. Therefore, most organizations maintain privacy policies, even though they are not required by law. Corporate privacy policies should address a customer's knowledge, control, notice and consent over the storage and use of information. They can also cover who has access to private data and when it can be used.

Multinational companies face an extremely difficult challenge in implementing data-collection and dissemination processes and policies because of the multitude of differing country or regional statutes. A good database design practice is to assign a single unique identifier to each customer – so that each has a single record describing all relationships with the company across all its business units. That way, the organization can apply customer privacy preferences consistently throughout all databases. Failure to do so can expose the organization to legal risks – aside from upsetting customers who opted out of some collection practices.

Right to Forget

In May 2014, an EU court ruling took a step towards giving people the 'right to be forgotten' by forcing Google and other search engines to remove certain links from search results. The content itself will still be available on the web, but it will just be more difficult to find. The implications of this have not yet sunk in, and Google's response has been to produce a form that people can use to request that search results be removed. Each request they receive will be assessed and a balance sought between the 'privacy rights of the individual and the public's right to know and distribute information'. It is thought that among those most likely to use the form will be people with spent convictions, victims of domestic violence and students wishing to tidy up their online image before submitting job applications.[18]

Individual Efforts to Protect Privacy

Many people are taking steps to increase their own privacy protection. Some of the steps that you can take to protect personal privacy include the following:

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- If you are concerned about what information a company is holding on you, use the Data Protection Act (or your country's equivalent) to find out what is stored about you in existing databases.
- Be careful when you share information about yourself. Don't share information unless it is absolutely necessary.
- Be vigilant in insisting that your doctor, bank or financial institution does not share information about you with others without your written consent.
- Be proactive to protect your privacy. For instance, you could get an unlisted phone number and think twice about registering for a service if it means you must supply a postal address. Consider registering for the telephone preference and mail preference services in your country (which stops commercial calls and post). In the UK the address is www.tpsonline.org.uk.
- When purchasing anything from a website, make sure that you safeguard your credit card numbers, passwords and personal information. Do not do business with a site unless you know that it handles credit card information securely (look for https:// in the address bar). Do not provide personal information without reviewing the site's data privacy policy.

When some people give out personal information, they change it slightly somehow, maybe changing their name from John T. Smith to John R. Smith. Then, in the future, if they get contacted as John R. Smith from an unknown source, they know which company the information must have come from and can take the appropriate steps.

13.5 Relevant Laws Governing Use of Technology

Many countries have laws that regulate privacy, data use and computer misuse. In Europe, the General Data Protection Regulation (GDPR) governs how companies, charities and other organizations can collect, use and share information, regardless of whether that information is stored electronically in a computer's memory, or on paper in a filing cabinet. The regulation applies equally to video data, including video that has been collected by CCTV in a shop, or on a body camera worn by the police. Introduced in 2016, the GDPR brought together all the existing but independent laws that individual European countries had enacted. It is based around seven data principles:

- Lawfulness, fairness and transparency
- Purpose limitation
- Data minimization
- Accuracy
- Storage limitation
- Integrity and confidentiality
- Accountability[19]

Going through these in order, they mean that: data that is collected and stored must be done so for a legal and publicly known purpose; that purpose must be specific – organizations are not supposed to grab data 'just in case' they can think of a purpose for it at a later date; organizations should not collect more data than they really need for their specified purpose; organizations must take steps to ensure the data is accurate; when the data is no longer needed it should be properly deleted; the data should be held securely; and organizations must be accountable – this means that if an organization stores data about you, you are entitled to 1) know about it,

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2) see a copy of the data that is held and 3) get that data corrected or deleted if there is a problem. 'Properly deleted' means that the data must be wiped from memory in a way that it is impossible to recover – something that simply pressing delete on your keyboard does not ensure.

Organizations of a certain size must appoint a data protection officer to ensure that these principles are adhered to. Other countries have similar legislation. South Africa has the Protection of Personal Information Act; Australia has a set of privacy acts.

Many countries also have laws that govern information held by public bodies. In the UK, the Freedom of Information Act 2000 allows anyone to see any information held by any public body. This includes government, councils, universities and hospitals, although there are exceptions to what information they have to hand over. The Act requires organizations to respond to any requests for information in a timely manner, either by supplying the information or stating why it cannot be supplied. The Act is a constant pain for many organizations who have to deal with time-wasting requests. However, many scandals of great public interest have been brought to light because of it. In the UK, one well-known example of this was the Members of Parliament expense claims (a lot of them were making frivolous claims), but the Act has also been used to investigate hospital waiting times, restaurant hygiene, council expenses and electoral fraud.[20, 21, 22]

The Computer Misuse Act is an older piece of legislation, coming into law in 1990 in the UK and essentially makes it illegal to use a computer to commit crime. It is the law that makes 'computer hacking' (gaining access to someone else's computer) an offence.

There are a number of more modern laws that govern our use of technology. One controversial example is the Regulation of Investigatory Powers Act 2000, which allows authorities to issue a 'disclosure notice' that essentially means people can be made to hand over the passwords to their phone or computer. London-based solicitors Sanders Law say that 'The police are able to request disclosure if the reason is to prevent or detect crime, if it's in the interests of national security or if it is in the interests of the economic wellbeing of the UK. This definition can be applied very widely to the extent that it can cover any crime, no matter how minor. Refusal to comply with a notice served under s49 of RIPA can result in a maximum sentence of two years imprisonment, or five years in cases involving national security or child indecency'.[23]

In 2018, Stephen Nicholson was jailed under the Regulation of Investigatory Powers Act for refusing to give police his Facebook password. In 2019, Nicholson was jailed for life for murdering a teenage girl. Note that to get the murder conviction, police did not in the end use his Facebook data.[24]

13.6 The Work Environment

The use of computer-based information systems has changed the make-up of the workforce. Jobs that require IS literacy have increased, and many less-skilled positions have been eliminated. Corporate programs, such as reengineering and continuous improvement, bring with them the concern that as business processes are restructured and information systems are integrated within them, the people involved in these processes will be removed.

However, the growing field of computer technology and information systems has opened up numerous avenues to professionals and nonprofessionals of all backgrounds. Enhanced telecommunications has been the impetus for new types of business and has created global markets in industries once limited to domestic markets. Even the simplest tasks have been aided by computers, making cash registers faster, smoothing order processing and allowing people with disabilities to participate more actively in the workforce. As computers and other IS components drop in cost and become easier to use, more workers will benefit from the increased productivity and efficiency provided by computers. However, information systems can raise other concerns.

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Health Concerns

Organizations can increase employee effectiveness by paying attention to the health concerns in today's work environment. For some people, working with computers can cause occupational stress. Anxieties about job insecurity, loss of control, incompetence and demotion are just a few of the fears workers might experience. In some cases, the stress can become so severe that workers might sabotage computer systems and equipment. Monitoring employee stress can alert companies to potential problems. Training and counselling can often help the employee and deter problems.

Computer use can affect physical health as well. Strains, sprains, tendonitis, tennis elbow, the inability to hold objects and sharp pain in the fingers can result. Also common is repetitive strain injury (RSI), including carpal tunnel syndrome (CTS), which is the aggravation of the pathway for nerves that travel through the wrist (the carpal tunnel). CTS involves wrist pain, a feeling of tingling and numbness, and difficulty grasping and holding objects. It can be caused by many factors, such as stress, lack of exercise and the repetitive motion of typing on a computer keyboard. Decisions on workers' compensation related to RSI have been made both for and against employees.

Other work-related health hazards involve emissions from improperly maintained and used equipment. Some studies show that poorly maintained laser printers can release ozone into the air; others dispute the claim. Numerous studies on the impact of emissions from display screens have also resulted in conflicting theories. Although some medical authorities believe that long-term exposure can cause cancer, studies are not conclusive at this time. In any case, many organizations are developing conservative and cautious policies.

Most computer manufacturers publish technical information on radiation emissions from their CRT monitors, and many companies pay close attention to this information. In addition, adjustable chairs and workstations should be supplied if employees request them.

In April 2014, French labour unions and employers in the high-tech and consulting field signed an agreement that employees must be able to disconnect communications tools. The agreement was widely reported as a ban on reading work-related emails after 6 pm although this is not the case. However, the move does show that the French recognize the negative impact that being always online can have on wellbeing.[25]

Avoiding Health and Environmental Problems

Many computer-related health problems are caused by a poorly designed work environment. The computer screen can be hard to read, with glare and poor contrast. Desks and chairs can also be uncomfortable. Keyboards and computer screens might be fixed in place or difficult to move. The hazardous activities associated with these unfavourable conditions are collectively referred to as 'work stressors'. Although these problems might not be of major concern to casual users of computer systems, continued stressors such as repetitive motion, awkward posture and eyestrain can cause more serious and long-term injuries. If nothing else, these problems can severely limit productivity and performance.

The science of designing machines, products and systems to maximize the safety, comfort and efficiency of the people who use them, called **ergonomics**, has suggested some approaches to reducing these health problems. The slope of the keyboard, the positioning and design of display screens, and the placement and design of computer tables and chairs have been carefully studied. Flexibility is a major component of ergonomics and an important feature of computer devices. People come in many sizes, have differing preferences and require different positioning of equipment for best results. Some people, for example, want to place the keyboard in their laps; others

ergonomics The science of designing machines, products and systems to maximize the safety, comfort and efficiency of the people who use them.

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prefer it on a solid table. Because of these individual differences, computer designers are attempting to develop systems that provide a great deal of flexibility. In fact, the revolutionary design of Apple's iMac computer came about through concerns for users' comfort, and after using basically the same keyboard design for over a decade, Microsoft introduced a new split keyboard called the Natural Ergonomic Keyboard 4000. The keyboard provides improved ergonomic features such as improved angles that reduce motion and how much you must stretch your fingers when you type. The design of the keyboard also provides more convenient wrist and arm postures, which make typing more convenient for users.[26]

Computer users who work at their machines for more than an hour per day should consider using LCD screens, which are much easier on your eyes than CRT screens. If you stare at a CRT screen all day long, your eye muscles can get fatigued from all the screen flicker and bright backlighting of the monitor. LCD screens provide a much better viewing experience for your eyes by virtually eliminating flicker and still being bright without harsh incandescence.[27, 28]

In addition to steps taken by hardware manufacturing companies, computer users must also take action to reduce strain injury and develop a better work environment. For example, when working at a workstation, the top of the monitor should be at or just below eye level. Your wrists and hands should be in line with your forearms, with your elbows close to your body and supported. Your lower back needs to be well supported. Your feet should be flat on the floor. Take an occasional break to get away from the keyboard and screen. Stand up and stretch while at your workplace. Do not ignore pain or discomfort. Many workers ignore early signs of strain injury, and, as a result, the problem becomes much worse and more difficult to treat.

Ethical and Societal Issues



Kettle Botnet Heats Up

In October 2016, a malware program called Mirai undertook a denial-of-service attack against a number of targets, including the popular cybersecurity blog KrebsOnSecurity. A denial-of-service attack involves flooding a website with millions of requests so that the genuine requests can't get through. Often these attacks originate from a botnet. Malware infects thousands of devices, and these devices – without their owner's knowledge – are then used to send out the denial-of-service requests. What is unusual about the Mirai malware is that rather than infecting what we traditionally think of as computers, it infected everyday devices such as kettles, video recorders and cameras that were connected to the Internet of Things.

Mirai tried various hardcoded root passwords on Internet-connected devices. Kettles weren't the most common object it attacked, but it was the one that created the most surprise among the general public. Here's what happened. Someone bought an Internet-enabled smart kettle. Makers advertise these devices as being able to save time for thirsty owners as they can remote boil from anywhere in the house, gain extra time in bed as they don't have to get up to switch the kettle on, boil water from the bus so it's ready to make tea as soon as the owner arrives home and save energy with ready notifications. The password that is used to gain access to the chips inside these are 'hardcoded'; in other words they cannot be changed by the owner. Mirai searched for devices, fed them the default password and then installed a program to make it join the botnet. When the botnet's owner wanted, they could command their army of kettles and other devices to start sending out messages to deny the service of the target website. Turning the kettle off and on would remove the malware, but this didn't help the target as most smart kettle owners didn't know what was going on. In addition, there is so much constant scanning going on for vulnerable devices that they can be re-infected within minutes of a reboot. Only changing the default password can prevent this.

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During the attack, connection problems began in the eastern USA before spreading to other parts of the USA and to Europe. Dozens of websites were at times unreachable including Mashable, CNN, the *New York Times*, the *Wall Street Journal*, Yelp and some businesses hosted by Amazon.

The *Atlantic* magazine then asked an interesting question: who is responsible if your kettle starts attacking Internet infrastructure? 'One idea would put more of a burden of legal responsibility on the manufacturers that produce connected devices – and hold them accountable if their products were implicated in cybercrime', said reporter Kaveh Waddell. Devices that are connected to the Internet of Things are often not designed with security in mind and this is a huge mistake. Holding manufacturers responsible may be the only way to change this. Mr Waddell acknowledges, however, that even if this started to happen tomorrow, 'the sea of poorly secured connected devices already out in the world will continue to haunt us for some time'.

KrebsOnSecurity reports that there are plenty of new, default-insecure Internet of Things devices being plugged into the Internet each day. Gartner Inc. forecasts that 20.8 billion connected things will be in use worldwide by the end of 2020.

Questions

1 What steps should manufacturers of Internet of Things devices now take to help deal with this problem?
2 Why would someone launch a denial-of-service attack?
3 Is a smart kettle a good idea?
4 What will have to happen to stop the 'sea of poorly secured connected devices' haunting us for years?

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13.7 Ethical Issues in Information Systems

As you've seen throughout the book in our Ethical and Societal Issues boxes, ethical issues deal with what is generally considered right or wrong. As we have seen, laws do not provide a complete guide to ethical behaviour. Just because an activity is defined as legal does not mean that it is ethical. As a result, practitioners in many professions subscribe to a **code of ethics** that states the principles and core values that are essential to their work and, therefore, govern their behaviour. The code can become a reference point for weighing what is legal and what is ethical. For example, doctors adhere to varying versions of the 2,000-year-old Hippocratic Oath, which medical schools offer as an affirmation to their graduating classes.

code of ethics A code that states the principles and core values that are essential to a set of people and, therefore, govern their behaviour.

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Some IS professionals believe that their field offers many opportunities for unethical behaviour. They also believe that unethical behaviour can be reduced by top-level managers developing, discussing and enforcing codes of ethics. Various IS-related organizations and associations promote ethically responsible use of information systems and have developed useful codes of ethics. The British Computer Society has a code of ethics and professional conduct that can be used to help guide the actions of IS professionals. These guidelines can also be used for those who employ or hire IS professionals to monitor and guide their work and can be seen at www.bcs.org. The international professional association ISACA (the Information Systems Audit and Control Association) which focuses on IT Governance, includes recognition of ethical issues in its governance framework COBIT (Control Objectives for Information and Related Technology). COBIT governs all aspects of IT development and management and is used to meet government regulations on reporting information, such as those set out for instance in the US Sarbanes-Oxley Act and the South African King III code of corporate governance. For more information see: www.isaca.org/COBIT.

The mishandling of the social issues discussed in this chapter – including waste and mistakes, crime, privacy, health and ethics – can devastate an organization. The prevention of these problems and recovery from them are important aspects of managing information and information systems as critical corporate assets. Increasingly, organizations are recognizing that people are the most important component of a computer-based information system and that long-term competitive advantage can be found in a well-trained, motivated and knowledgeable workforce.

Summary

Policies and procedures must be established to avoid computer waste and mistakes. Computer waste is the inappropriate use of computer technology and resources in both the public and private sectors. Computer mistakes relate to errors, failures and other problems that result in output that is incorrect and without value. Waste and mistakes occur in government agencies as well as corporations. At the corporate level, computer waste and mistakes impose unnecessarily high costs for an information system and drag down profits. Waste often results from poor integration of IS components, leading to duplication of efforts and overcapacity. Inefficient procedures also waste IS resources, as do thoughtless disposal of useful resources and misuse of computer time for games and personal processing jobs. Inappropriate processing instructions, inaccurate data entry, mishandling of IS output and poor systems design all cause computer mistakes.

A less dramatic, yet still relevant, example of waste is the amount of company time and money employees can waste playing computer games, sending unimportant email or accessing the Internet. Junk email, also called spam, and junk faxes also cause waste.

Preventing waste and mistakes involves establishing, implementing, monitoring and reviewing effective policies and procedures. Careful programming practices, thorough testing, flexible network interconnections and rigorous back-up procedures can help an information system prevent and recover from many kinds of mistakes. Companies should develop manuals and training programmes to avoid waste and mistakes. Company policies should specify criteria for new resource purchases and user-developed processing tools to help guard against waste and mistakes.

Computer crime is a serious and rapidly growing area of concern requiring management attention. Some crimes use computers as tools (e.g. to manipulate records, counterfeit money and documents, commit fraud via telecommunications links and make unauthorized electronic transfers of money). Identity theft is a crime in which an imposter obtains key pieces of personal identification information to impersonate someone else. The information is then used to obtain credit, merchandise and services in the name of the victim, or to provide the thief with false credentials.

A cyberterrorist is someone who intimidates or coerces a government or organization to advance his or her political or social objectives by launching computer-based attacks against computers, networks and the

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information stored on them. A cracker, or criminal hacker, is a computer-savvy person who attempts to gain unauthorized access to computer systems to steal passwords, corrupt files and programs, and even transfer money. Script kiddies are crackers with little technical savvy. Insiders are employees, disgruntled or otherwise, working solo or in concert with outsiders to compromise corporate systems.

Computer crimes target computer systems and include illegal access to computer systems, alteration and destruction of data and programs by viruses (system, application and document), and simple theft of computer resources. A virus is a program that attaches itself to other programs. A worm functions as an independent program, replicating its own program files until it destroys other systems and programs or interrupts the operation of computer systems and networks. Malware is a general term for software that is harmful or destructive. A Trojan horse program is a malicious program that disguises itself as a useful application and purposefully does something the user does not expect. A logic bomb is designed to 'explode' or execute at a specified time and date.

Because of increased computer use, greater emphasis is placed on the prevention and detection of computer crime. Antivirus software is used to detect the presence of viruses, worms and logic bombs. Use of an intrusion detection system (IDS) provides another layer of protection in the event that an intruder gets past the outer security layers – passwords, security procedures and corporate firewall. It monitors system and network resources and notifies network security personnel when it senses a possible intrusion. Many SME organizations are outsourcing their network security operations to managed security service providers (MSSPs), which monitor, manage and maintain network security hardware and software.

Software and Internet piracy might represent the most common computer crime. Computer scams have cost people and companies thousands of euros. Computer crime is also an international issue.

Many organizations and people help prevent computer crime. Security measures, such as using passwords, identification numbers and data encryption, help to guard against illegal computer access, especially when supported by effective control procedures. Public-key infrastructure (PKI) enables users of an unsecured public network such as the Internet to securely and privately exchange data through the use of a public and a private cryptographic key pair that is obtained and shared through a trusted authority. The use of biometrics, involving the measurement of a person's unique characteristics, such as the iris, retina or voice pattern, is another way to protect important data and information systems. Virus-scanning software identifies and removes damaging computer programs. Although most companies use data files for legitimate, justifiable purposes, opportunities for invasion of privacy abound. Privacy issues are a concern with government agencies, email use, corporations and the Internet. A business should develop a clear and thorough policy about privacy rights for customers, including database access. That policy should also address the rights of employees, including electronic monitoring systems and email. Fairness in information use for privacy rights emphasizes knowledge, control, notice and consent for people profiled in databases. People should know about the data that is stored about them and be able to correct errors in corporate database systems. If information on people is to be used for other purposes, they should be asked to give their consent beforehand. Each person has the right to know and the ability to decide. Platform for Privacy Preferences (P3P) is a screening technology that shields users from websites that don't provide the level of privacy protection they desire.

Jobs, equipment and working conditions must be designed to avoid negative health effects. Computers have changed the make-up of the workforce and even eliminated some jobs, but they have also expanded and enriched employment opportuntities in many ways. Computers and related devices can affect employees' emotional and physical health. Some critics blame computer systems for emissions of ozone and electromagnetic radiation.

The study of designing and positioning computer equipment, called ergonomics, has suggested some approaches to reducing these health problems. Ergonomic design principles help to reduce harmful effects and increase the efficiency of an information system. The slope of the keyboard, the positioning and design of display screens, and the placement and design of computer tables and chairs are essential for good health. Good practice includes keeping good posture, not ignoring pain or problems, performing stretching and strengthening exercises, and seeking proper treatment. Although they can cause negative health consequences, information systems can also be used to provide a wealth of information on health topics through the Internet and other sources.

Ethics determine generally accepted and discouraged activities within a company and society at large. Ethical computer users define acceptable practices more strictly than just refraining from committing crimes; they also consider the effects of their IS activities, including Internet usage, on other people and organizations. Many IS professionals join computer-related associations and agree to abide by detailed ethical codes.

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Self-Assessment Test

1 Fooling or convincing someone into giving you their password is called _____.

2 Another name for a zombie network is _____.

3 _____ software runs in the background protecting your computer from infected files.

4 _____ are people who illegally duplicate software.

5 A _____ is a slang term for someone paid by a company to make them look good and make you a victim of a scam.

6 _____ allows you to send information securely through a public network.

7 _____ involves identifying you from your physical traits.

8 _____ watches for people illegally using a company's network.

9 Your work email account is legally accessible only by you. True or false?

10 Under the _____ an individual can ask for certain web pages to be removed from the results of web searches.

Review Questions

1 Identify several potential sources of computer-related waste.

2 List several examples of computer-related mistakes.

3 Give some examples of computer crime.

4 What is identity theft?

5 Who can prevent computer crime?

6 What is social engineering?

7 What is a code of ethics? Give an example.

8 How can you protect yourself from identity theft?

9 Give an example of biometrics.

10 How can you create a secure password?

Discussion Questions

1 How might you guess someone's password?

2 Why do you think people write computer viruses?

Web Exercises

1 Search for information about cyber warfare. Has there ever been a cyber war?

2 What information can you find out about the dean of your faculty? What additional information could you get through a freedom of information request?

Case One

Open-Access Pirates

The world of research publishing is a strange place. Scholars are often employed by universities to create (research) and share (teach) knowledge. Often they are paid – at least in part – by the tax payer. The knowledge that they create is published in a number of places including academic journals. Scholars write the papers which are then reviewed and edited by other scholars. None of this is paid for by the journal's publisher. The publisher then publishes the journal and sells it back to university libraries who often subscribe to many hundreds such journals. Many see the fees that they charge as being so high they

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are obscene. Tax payers, most of whom do not have access to university libraries and who paid for the research in the first place, have to purchase access to individual papers for over €30 if they want to read them. Scholars whose libraries don't subscribe are expected to pay this as well, although in fact most authors would just send a copy of their paper to anyone who asked. Many see this system as being unfair and exploitative. This is certainly the view of Alexandra Elbakyan.

Elbakyan is seen as a hero or a villain, depending on who you talk to. In 2011, the graduate student from Kazakhstan opened a searchable online database giving free access to 50 million journal articles. The technical details of how the system works have not been fully published, but it looks like it relies on library login details that have been donated by sympathetic university employees, and these are used to access the papers.

'There are many ways to argue that copyright infringement is not theft, but even if it is, it is justified in this case', she told the *Washington Post* from her place in hiding somewhere in Eastern Europe or Russia. 'All content should be copied without restriction. But for education and research, copyright laws are especially damaging.' It would seem that many agree with her. *Science* magazine has said that her site, Sci-Hub, has millions of users. Publishers are not impressed. They have slapped lawsuits on the site but have been unable to completely shut it down. Elsevier brought a copyright lawsuit against it in 2015 in a New York court, claiming lost revenue, but as Elbakyan does not live in the USA, is not a US citizen and has no assets in the USA, not much can really be done to stop her by that court.

Science magazine was given some data from Elbakyan on the use of Sci-Hub. These data revealed that it has become the world's de facto open-access research library. Sci-Hub users are not limited to the developing world, as some have thought. In fact, many scholars seem to be using the site for convenience rather than necessity. A full quarter of Sci-Hub downloads came from the 34 members of the Organization for Economic Cooperation and Development, the wealthiest nations with presumably the best journal access. In fact, some of the most intense use of Sci-Hub appears to be happening on the campuses of US and European universities.

Elbakyan has now enrolled in a history of science master's programme at a small private university in an undisclosed location. Appropriately enough, her thesis focuses on scientific communication. 'I perceive Sci-Hub as a practical side of my research.'

Questions

1. In what ways might Sci-Hub disrupt traditional journal publishing?
2. How can international websites be policed?
3. Why are researchers at well-resourced universities using Sci-Hub?
4. What lessons can all publishers learn from Sci-Hub?

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Case Two

The Encryption Row Is Back

In 2015, then UK prime minister David Cameron said, 'Are we going to allow a means of communication where it simply isn't possible to [read electronic communications even under a warrant from the Home Secretary] and my answer to that question is no we must not'. He was talking about encrypted communications, and as we have seen earlier in this text (see Case One in Chapter 6) with some instant messaging services that use end-to-end encryption, it isn't possible for the company running the service to hand over messages to the government or anyone else as they simply don't have access to them!

In response to calls like these from politicians, a group of computer security experts published a paper called 'Keys Under Doormats: Mandating Insecurity by Requiring Government Access to All Data and Communications' which looked at the technical problems in allowing access to two types of information: texts sent via instant messaging services and information stored on an encrypted mobile phone.

Briefly, encrypted instant messages are encrypted with a throw-away key that not even the instant message company has access to – it is only available to the two devices that are communicating with each other. In order to make messages available to governments and law enforcement agencies, the suggestion is that the throw-away key itself be encrypted with a government key and sent alongside the message. A recording of the message could then be decoded by the government – they would first decrypt the throw-away key and then use it to decrypt the message.

The implication of this is that the throw-away key would no longer be throw-away. In fact it would be forever stored in encrypted form, ready to be used when needed. It also means that all messages would be kept safe by one government-owned key. If – or rather, according to many experts, when – that key is leaked, all messages would suddenly be available.

Alternatively the government could ban the use of encryption by instant messaging companies. One problem with this is that it involves forcing a company in one country to obey the law in another country. A second problem is that encryption algorithms are simple and well known and could easily be implemented by someone with a little programming knowledge. We also should note that all Internet banking and e-commerce relies on exactly the same techniques as encrypted instant messaging – if you ban them then you would no longer be able to securely text chat with live help from your bank.

The second scenario, gaining access to data stored on encrypted mobile phones, has helped in numerous court cases, but it does involve the active help of the tech company involved (Apple or Google, for instance). The current position is that many modern encrypted phones cannot be opened, even by the tech firms.

The security experts argue that anything that weakens security for some (those involved in court cases) weakens security for all, and that this should be avoided. Very similar arguments were made in the 1990s when encryption techniques were classed by the US Army as a munition and subject to the same laws as guns, bullets and missiles. At that time, those arguing against banning encryption won their battle. It would seem that they are about to win again if for no other reason than once the techniques are out of the bag, it is impossible to put them back in. This will likely remain the case until quantum computers (which would make current encryption techniques insecure) are perfected!

Questions

1. Should instant messages be available for law enforcement to read?
2. How could a government encryption be leaked?
3. Might mobile phone owners not benefit from the tech company being able to get inside them, if for instance they forgot their passcode?
4. If you represented Google and you were refusing to provide access to a phone that may contain evidence of a crime, what would you say to the victim of that crime?

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Case Three

On Patrol With the Love Police

There are many communities within the video sharing site YouTube that specialize in sharing videos which centre around certain themes. We have already discussed 'let's play' videos where vloggers (video bloggers) share footage of themselves playing computer games. There is another, much smaller community that makes videos about what we might call 'authority baiting'. A typical scene is as follows. A camera is set up on a high street pointing into a bank or shop. Inevitably, a security guard takes an interest and approaches the vlogger. The guard then verifies that filming is taking place. Once this is confirmed, they tell the vlogger that 'You cannot film here' giving the opportunity for an argument. This can make for entertaining footage, especially if the security guard responds aggressively. Many variations can be found including filming bailiffs, traffic wardens, university campus security, shopping centre security, police and postal workers. The vlogger is usually arguing from a better informed position than the security guard with a working knowledge of relevant laws and cases, backed up with a moral argument about the number of CCTV cameras filming inside the bank and around the city centre. On at least one occasion the vlogger announces Action!' to the camera as the guard approaches!

In the late 1990s, a group of technology experts led by Steve Mann created wearable technology that could record everything that the wearer saw and did (see Chapter 10 for more about this). Back then they were already talking about smart floors, smart elevators and smart light switches that could be used for ubiquitous surveillance inside buildings. With the popularity of smartphones, tablet computers, other mobile devices and even smart TVs, the opportunity for, and the practice of, surveillance inside what were once very private spaces is now very real. The imbalance between those surveying and the inability of those under surveillance to control it has created what, towards the end of the 18th century, Jeremy Bentham imagined as a *panopticon*. The concept referred to the ability of a single watchman to observe all inmates of an institution without them being able to tell whether or not they were being watched. In response to what he saw as the emergence of the panopticon, Steve Mann and others began to confront surveillance with their technological creations, many of which were wearable cameras. These were used to 'record the recorders' and essentially asked organizations: *how do you like it?* The term they coined for this was sousveillance, and indeed the concept captures closely what the YouTube authority baiters are doing. They describe sousveillance as a performance that uses recording technology to:

a) uncover the panopticon and undercut its primacy and privilege
b) relocate the relationship of the surveillance society within a more traditional commons notion of observability.

Today, authority baiters on YouTube seek to highlight the hypocrisy of filming the public but not allowing filming by the public even though in many countries the law is clear – there can be no expectation of privacy on public land. London Metropolitan Police, for example, state on their website that 'members of the public and the media do not need a permit to film or photograph in public places and police have no power to stop them filming or photographing incidents or police personnel'.

The Love Police is a group of authority baiters. They describe themselves as an 'absurdist performance art group which concentrates on issues such as the encroaching fear and greed in our society'. The group would regularly hold up signs stating 'Everything is OK' claiming they were trying to make people realize that everything is not OK: authority figures regularly try to prevent citizens enjoying freedoms to behave in ways that they want to reserve for themselves, such as recording in public spaces and limiting free expression in public spaces. The group say they do not take themselves too seriously and that they try to see the human under the uniform. Indeed, often they offer to hug

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that human. They say that the best footage they have is when the 'facade of the training and the indoctrination of a police officer break[s] down and he looks at you as a fellow human being'.

Questions

1. Why do you think authority baiters film themselves and share their videos?
2. Would authority baiting exist without *video-sharing* sites like YouTube?
3. Should we object to being filmed in city centres and inside stores?
4. Can YouTube lead to social change?

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World Views Case

Facebook: A Platform for Cyberbullying and Cyber Racism or Not?

By Natasha Madhav
IIE, South Africa

Social media can be described as an innovative tool that has increased collaboration and engagement among people; however, recent trends in social media have seen its contribution to cyberstalking, cyberbullying, racism and hacking into accounts to provide potential threats to social media users. The web is progressively being used by people to attack, bully and circulate racist ideologies, as well as to communicate, organize and mobilize. Cyberbullying is bullying that happens across an online medium generally using devices such as mobile phones, personal computers and tablets as well as online communication mediums such as social media sites, SMSs, online chats and websites.

There are various forms of racist and bullying activity on the web. These include websites, social networks, computer games and online chat-rooms. Twitter and Facebook, the most popular social media sites, have seen their share of cyberstalking, racist attacks on communities as well as cyberbullying.

Cyberbullying among Aboriginal people in Australia has become a perturbing trend. In 2012, Aboriginal people called for a ban on social media including Facebook as it was perceived to fuel tension and hate speech among communities. While some communities were requesting a ban on Facebook and other sites, other members felt that Facebook could provide an educational platform for Aboriginal people. The other concern participants voiced about Facebook and suicidal behaviour was in regards to suicide clusters. Research conducted on Australian Aboriginal communities indicated that their suicide rates are among the highest in the world. Research conducted by *The Elders Report* (2014) indicates that suicide among Aboriginal people is 40 per cent higher than the general Australian population, making them among the highest in the world.

Further research funded by the Australian Research Council reveals how their engagement on social media has assisted the strategies aimed at preventing suicide. Facebook, for instance, has been able to provide a platform for the Aboriginals as they seek advice and support for issues relating to suicide. Increased use of mobile technologies has also been a significant contributor to Aboriginal youth and their engagement on Facebook.

Research conducted by Carlson (2013) describes how social media is changing the way we communicate and connect with each other globally. For Aboriginals, Facebook and other social media provide a point of connection allowing people also using these channels to share their thoughts, concerns and the cultural activities in their communities. Carlson (2013) further states that the Aboriginal people are also participating in political and cultural activities on social media, which can be the platform for notifications of deaths and extending support to the grieving. They also revealed that social media can be both a productive and problematic space for Aboriginal people. While social media provided an online space where families could assemble, it also created a space where cultural compassion, values and performances are displayed publicly, thus allowing the Aboriginal people to become subject to abuse, racism and hate speech. Australia's Race Discrimination Commissioner has also raised the issue of cyber racism, stating that there has been an increase in complaints about online racism to the extent where Georgatos (2014) argued that racism had contributed to some suicides in Aboriginal communities.

On a brighter side, the Aboriginal People describe Facebook as a site that helped them to engage with family and friends and identify people at risk and who needed help. This also enabled quicker responses to get help by calling emergency or other services. Another important aspect was their ability to share their thoughts and concerns about their communities. So it seems that even though Facebook and Twitter encourage bullying, racism and suicidal talk, they also have the ability to reduce suicidal tendencies, create an online community and serve as a platform for prevention interventions.

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Questions

1. What do you understand by the term cyber racism? Could you provide any recent examples in your communities?
2. Do you think that social media such as Facebook provides a space of support and care for the Aboriginal people?
3. What potential barriers might the Aboriginal people have experienced using these tools?
4. Do you think Facebook has the potential to help people from various communities to overcome barriers, such as personal, social and cultural, or is it likely to facilitate racial abuse and social segregation?

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